

DETERMINISTIC PATH ENSEMBLES, DRIVEN COLLISION MAPS, AND ENDOGENOUS CONJUGATE PARAMETERS

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ABSTRACT. We construct a single finite-dimensional autonomous hybrid Hamiltonian realization of the driven baker benchmark and separate it from its symbolic natural extension. Parameter derivatives are defined on a fixed Sobolev–conormal Hilbert scale carrying bulk, face, corner, and higher flag currents. A closed all-order chain rule yields explicit response words with uniform remainder bounds. A Hilbert-valued Gordin decomposition then gives a Radon current-valued functional central limit theorem with trace-class covariance. Mechanical work and risk calibration remain distinct typed claims.

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1. ROUND-TWENTY-ONE POSITIVE CLOSURE: PROPER SUSPENSION, INCIDENCE-DUAL CURRENTS, AND A PROVED HILBERT FCLT

This section replaces the former Round-Seventeen module. The parameter family is now part of the mathematical data rather than an assumption

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introduced in a proof: the seams $s_i(a)$ are C^∞ on a compact interval K , and $w_i(a) = s_i(a) - s_{i-1}(a) \geq w_* > 0$. Finite-order statements require only the corresponding finite number of derivatives. The construction below never restricts an arbitrary negative-Sobolev distribution to a face. All traces are taken on positive-regularity test jets and are then dualized.

1.1. The labelled section and its proper mapping torus. Let $\widehat{S}_a$ be the disjoint union of the closures of the rectangles $S_{ijk}(a)$, with every oriented face and quadrant germ retained as part of the label. Thus two geometric points on a seam are distinct whenever their one-sided germs are distinct. On the component with active branch j put

$$F_a(q, p; i, j, k) = \left(\frac{q - s_{j-1}}{w_j}, s_{j-1} + w_j p; j, k, \ell \right),$$

where the oriented germ of the first coordinate determines ℓ . The horizontal target label records j , so the inverse branch is unique even on a labelled face.

Lemma 1.1 (Labelled baker diffeomorphism). *The map $F_a : \widehat{S}_a \rightarrow \widehat{S}_a$ is a C^∞ diffeomorphism of a finite compact manifold with corners, stratum by stratum. It preserves $dq \wedge dp$ and satisfies*

$$F_a^*(p dq) - p dq = dG_a$$

for a smooth function G_a on the labelled section. The family $a \mapsto (F_a, G_a)$ is C^∞ in fixed affine cell charts.

Proof. On each labelled rectangle the derivative is $\text{diag}(w_j^{-1}, w_j)$ and has determinant one. The target horizontal germ is the old branch j ; together with the target vertical germ it determines the unique inverse formula $q = s_{j-1} + w_j Q$ and $p = (P - s_{j-1})/w_j$. Incidence maps between labelled faces are affine restrictions of these formulas, so the cellwise maps paste as a diffeomorphism of the labelled corner complex. Direct calculation gives

$$F_a^*(P dQ) - p dq = \frac{s_{j-1}}{w_j} dq = d \left[\frac{s_{j-1}}{w_j} (q - s_{j-1}) \right]$$

on a branch. Add constants, determined successively on the finite incidence graph, so that the primitives agree on every labelled common face. There is no compatibility cycle obstruction because the sum of primitive differences around a cycle is the primitive of the identity map and is therefore constant; the constants remove it. Smooth parameter dependence follows in the fixed coordinates $u = (q - s_{j-1})/w_j$, $v = (p - s_{i-1})/w_i$. $\square$

Define the mapping torus

$$Q_a = (\widehat{S}_a \times [0, 1]) / ((x, 1) \sim (F_a x, 0)).$$

Choose $\chi \in C^\infty([0, 1])$ equal to zero near 0 and one near 1. On the cylinder put

$$\Lambda_a = p dq + d(\chi(t)G_a), \quad \Theta_a = \Lambda_a - E dt, \quad \Omega_a = d\Theta_a.$$

The trace at $t = 1$ is F_a^* of the trace at $t = 0$.

Theorem 1.2 (Proper exact Hamiltonian suspension). *The space Q_a is a Hausdorff, second-countable smooth manifold with corners. The form Ω_a descends to an exact symplectic form on $\mathcal{M}_a = Q_a \times \mathbb{R}_E$. For the convention $\iota_{X_H} \Omega_a = dH$, the Hamiltonian $H = E$ has $\dot{t} = 1$, is complete, and has first return time one and first return map F_a on the section $t = 0$. Forgetting the labels factors this return to the ordinary baker map outside the null union of seam iterates. No additional return occurs.*

Proof. For a compact Hausdorff space and a homeomorphism, the equivalence relation used in the mapping torus is closed: away from the two boundary collars it is the diagonal, and on the collars it is the graph of F_a or F_a^{-1} . It is also proper because its equivalence classes have at most two cylinder representatives. Hence the quotient is Hausdorff and second countable. Choose corner collars $U \times [0, \varepsilon)$ at the two cylinder ends. The map $(x, r) \mapsto (F_a x, r)$ identifies the collars and supplies the quotient charts; compatibility follows from the ordinary cocycle identity for the single diffeomorphism F_a . This proves the smooth corner structure rather than inferring it from separate face graphs.

The boundary equality $\Lambda_a|_{t=1} = F_a^*(\Lambda_a|_{t=0})$ proves that Λ_a descends. Consequently so does Θ_a , and

$$\Omega_a = d(p dq) + dt \wedge dE$$

in every cylinder chart. It is closed and nondegenerate, including on each corner stratum. Since $\iota_{\partial_t} \Omega_a = dE$, the Hamiltonian vector field of E is ∂_t . Its orbit crosses the section exactly when the clock passes an integer and the mapping-torus identification then applies F_a . Completeness follows from the complete translation flow in the clock and the compactness of the section. $\square$

Adjoin a work pair (z, π_z) and replace Θ_a by $\Theta_a - \pi_z dz$. In a clock window disjoint from the gluing collars add $\pi_z c_j \rho(t)$, where $\int_0^1 \rho = 1$ and j is the retained branch label. The resulting globally smooth Hamiltonian translates z by c_j on that return and leaves (q, p) unchanged.

1.2. An incidence-completed positive test scale. Let $\mathcal{F}_n(a_0)$ be the finite family of all labelled bulk, face, and corner strata obtained from forward and backward seam preimages of depth at most n at a fixed reference parameter a_0 . A fixed affine transport $U_{F,a} : F(a_0) \rightarrow F(a)$ identifies every stratum throughout K ; at a birth the two oriented germs are retained separately, so no dimension-changing identification is made. There are constants C_0, d_0 with $\#\mathcal{F}_n \leq C_0 d_0^n$.

Fix $0 < \beta < d_0^{-2}$, integers $q \geq 0$ and $m > 2q + 6$, and write $c(F)$ for codimension. Define the Hilbert test-jet space

$$\Phi_q^m = \bigoplus_{n \geq 0} \bigoplus_{F \in \mathcal{F}_n} \bigoplus_{0 \leq r \leq q - c(F)} \beta^{-n/2} H^{m+q+2-c(F)-r}(F; N_F^{\otimes r}).$$

Repeated occurrences of the same geometric set retain their labels. A subdivision is represented by the orthogonal sum of its children, with the parent component identified with their restriction-compatible diagonal. The closed span of these diagonal relations is denoted $\mathcal{R}_q^m$. Put

$$\widehat{\Phi}_q^m = \Phi_q^m / \mathcal{R}_q^m, \quad \mathbb{J}_q^m = (\widehat{\Phi}_q^m)'.$$

Thus a current is a continuous functional on positive-regularity labelled jets. It is not an arbitrary ambient element of H^{-m} .

Theorem 1.3 (Typed incidence calculus). *For every q the following maps are bounded on the displayed scale.*

- (i) *Restriction and normal differentiation of a test component to a child stratum, by the ordinary positive Sobolev trace theorem.*
- (ii) *The adjoint extension of a child current to its parent and the signed cellular boundary map. These are the only maps called face or corner maps on currents.*
- (iii) *Pullback by F_a on tests and hence push-forward by F_a on currents. A depth shift has norm at most $C\sqrt{\beta}$ and is summable.*
- (iv) *Refinement and coarsening maps; they are mutually inverse isometries on the quotient and preserve every label not removed by an internal-face relation.*

The spaces are separable Hilbert spaces, and distinct labelled currents cannot cancel merely because their ambient distributions agree.

Proof. All test exponents exceed the relevant codimension divided by two plus the normal order, so the positive trace theorem applies before dualization. Its adjoint is an extension map, exactly the direction supplied by duality. The finite incidence matrices are triangular and have uniformly bounded entries in the fixed affine charts. A branch changes depth by at most one; after squaring the norm, its combinatorial multiplicity is bounded by d_0 and its weight gain is β , so $d_0\beta < 1$. The diagonal subdivision relations are closed because restriction is bounded. Quotients of Hilbert spaces by closed subspaces are Hilbert, and the countable direct sum is separable. Orthogonal labels remain different coordinates until an explicit internal-face relation identifies them, so ambient cancellation is impossible. $\square$

Let $V_{F,a} = \partial_a U_{F,a} \circ U_{F,a}^{-1}$. On the smooth test core define

$$\mathcal{A}_a \phi = \partial_a \phi + \mathcal{L}_{V_{F,a}} \phi$$

componentwise, including the induced normal-bundle connection. Its transpose defines the material derivative

$$\langle \mathfrak{D}_a J, \phi \rangle = -\langle J, \mathcal{A}_a \phi \rangle.$$

Proposition 1.4 (Fixed-fibre material derivative). *For every q the operator $\mathfrak{D}_a$ is bounded from $\mathbb{J}_{q+1}^{m+2}$ to $\mathbb{J}_q^m$, uniformly for $a \in K$, and is therefore closed*

after taking the graph closure of the smooth core. Moreover

$$\|\partial_a^r(F_a)_\# J\|_{\mathbb{J}_q^m} \leq C_q^{r+1} r! \|J\|_{\mathbb{J}_{q+r}^{m+2r}}$$

for an analytic seam family, and with $r!$ replaced by the corresponding C^r seminorm for a merely C^r family.

Proof. In the fixed charts, $\mathcal{A}_a$ is a first-order differential operator with smooth bounded coefficients and one bounded triangular incidence term. It maps $\widehat{\Phi}_q^m$ continuously into $\widehat{\Phi}_{q+1}^{m+2}$; transposition gives the stated bounded map in the opposite direction. Hence if $J_n \rightarrow 0$ in the domain space and $\mathfrak{D}_a J_n \rightarrow K$ in the target, then for every test jet ϕ ,

$$\langle K, \phi \rangle = \lim_n -\langle J_n, \mathcal{A}_a \phi \rangle = 0,$$

so $K = 0$. This is the fixed-fibre graph criterion for closability. Repeated differentiation gives finite Faà di Bruno sums of coefficient derivatives, Lie derivatives, and bounded incidence maps. Cauchy estimates for an analytic family give the factorial bound; the depth weight controls every preimage shift. $\square$

1.3. All-order response on the coupled Bernoulli extension. Let $\Sigma = \{1, 2, 3, 4\}^{\mathbb{Z}}$ with product law $\mu_a = \bigotimes_{k \in \mathbb{Z}} (w_1(a), \dots, w_4(a))$ and shift σ . The coding of the regular baker section is a measure isomorphism. On future and past one-sided Hölder spaces define

$$P_a^+ f(x) = \sum_i w_i(a) f(ix), \quad P_a^- g(y) = \sum_i w_i(a) g(iy).$$

Their centered powers contract by ρ^n for one common $\rho < 1$. Denote the reduced resolvents by $R_a^\pm$. A mixed cylinder is conditioned on its present symbol; this finite-rank conditioning operator is denoted C_a . Scores are the centered multipliers $S_a^{(r)}(i) = \partial_a^r \log w_i(a)$ with their lower-order Bell corrections.

For a rooted ordered tree T with total derivative order r , label every vertex by one of: a material derivative $\mathfrak{D}_a^{(j)}$, a future score $S_a^{(j)} R_a^+$, a past score $S_a^{(j)} R_a^-$, or a derivative of the finite present coupling C_a . Let $\mathcal{T}_r$ be the finite set of such trees and $b(T)$ its ordinary Faà di Bruno coefficient.

Theorem 1.5 (Convergent response tree formula). *If $J_a \in \mathbb{J}_{r+1}^{m+2r+2}$ is a C^{r+1} transported current, then*

$$\frac{d^r}{da^r} \int J_a d\mu_a = \sum_{T \in \mathcal{T}_r} b(T) \Lambda_a(\mathcal{O}_T J_a),$$

where every composition $\mathcal{O}_T$ is typed by Theorem 1.3 and Proposition 1.4. The sum is the limit of the exact finite-cylinder derivative and obeys

$$\sum_{T \in \mathcal{T}_r} |b(T)| \|\Lambda_a \mathcal{O}_T\| \leq C_r (1 - \rho)^{-2r-2}.$$

It is invariant under dynamic refinement. For a fixed physical observable, the material, score, and incidence contributions sum to zero, as required by Lebesgue invariance.

Proof. For a cylinder depending on coordinates $[-N, N]$, differentiation of the finite product density and of the transported current is ordinary finite dimensional calculus. Center every score. Successively condition the future and past tails and solve the two Poisson equations; this inserts R_a^+ and R_a^- . Conditioning at coordinate zero gives C_a . Repeating Leibniz' rule produces exactly the ordered trees and their Bell coefficients. The resolvents have norm at most $(1 - \rho)^{-1}$, and the material and incidence operators have the bounds proved above.

If $J_a^{(N)}$ is the conditional cylinder approximation, the Bernoulli tail and the depth weight give $\|J_a - J_a^{(N)}\| \leq C(\sqrt{d_0}\beta)^N$, uniformly with all derivatives through order r . Hence the finite formulas converge absolutely and may be differentiated term by term. Refinement changes a presentation by an element of $\mathcal{R}_q^m$, annihilated by every quotient functional. Finally, for a fixed physical observable, the exact change-of-variables identity $\int F \circ F_a dqdp = \int F dqdp$ holds for every a ; differentiating the already justified response formula yields the stated cancellation. $\square$

1.4. The Hilbert-valued projective estimate and FCLT. Let $Y_k^{(q)}$ be the centered order- q current increment, coded on Σ and taking values in $\mathbb{J}_q^m$. Let $\mathcal{F}_j = \sigma(x_i : i \leq j)$. The depth- n part of $Y_0^{(q)}$ depends only on symbols in $[-n, n]$ and its squared norm is bounded by $C(d_0\beta)^n$.

Lemma 1.6 (Explicit projective summability). *There are $C < \infty$ and $0 < \vartheta < 1$ such that*

$$\|\mathbb{E}(Y_0^{(q)} \mid \mathcal{F}_{-n}^0) - \mathbb{E}(Y_0^{(q)} \mid \mathcal{F}_{-n+1}^0)\|_{L^2(\mathbb{J}_q^m)} \leq C\vartheta^n.$$

The same estimate holds for the future filtration. Consequently the two projective series and the series of trace norms of covariance operators are summable.

Proof. Conditional expectations cancel all components whose oldest required symbol is earlier than $-n$. What remains is the orthogonal sum of strata of depth at least n . Its squared L^2 norm is at most $C \sum_{r \geq n} (d_0\beta)^r$. Take $\vartheta = \sqrt{d_0}\beta < 1$. For covariance, write for $k > 0$

$$\|\mathbb{E}(Y_0 \otimes Y_k)\|_1 \leq \|Y_0\|_{L^2} \|\mathbb{E}(Y_k \mid \mathcal{F}_0)\|_{L^2} \leq C\vartheta^k,$$

using the rank-one trace norm $\|u \otimes v\|_1 = \|u\| \|v\|$ and the future projective estimate. Negative k is identical. $\square$

Theorem 1.7 (Radon current-valued functional central limit theorem). *The series*

$$\Sigma_q = \sum_{k \in \mathbb{Z}} \mathbb{E}(Y_0^{(q)} \otimes Y_k^{(q)})$$

converges absolutely in trace norm and defines a positive trace-class operator. Moreover

$$n^{-1/2} \sum_{j < nt} Y_j^{(q)} \implies W_q(t)$$

in $D([0, 1], \mathbb{J}_q^m)$, where W_q is the Radon Brownian motion with covariance Σ_q .

Proof. Lemma 1.6 gives the Hilbert Gordin decomposition $Y_j = M_j + H \circ \sigma^{j+1} - H \circ \sigma^j$, with M_j a stationary square-integrable martingale difference and $H \in L^2(\mathbb{J}_q^m)$. The conditional bracket averages converge in trace norm by the ergodic theorem applied first to finite-rank projections and then using the summable trace bound. The Hilbert martingale invariance principle follows from the scalar martingale theorem on every finite-rank projection, the Lindeberg estimate

$$\mathbb{E}[\|M_0\|^2 \mathbf{1}_{\{\|M_0\| > \varepsilon \sqrt{n}\}}] \longrightarrow 0,$$

and compact containment obtained from the trace tails of the bracket.

For the coboundary,

$$\max_{j \leq n} \|H \circ \sigma^j\| / \sqrt{n} \longrightarrow 0$$

in probability because $H \in L^2$ implies $n \mathbb{P}(\|H\| > \varepsilon \sqrt{n}) \rightarrow 0$. Hence its two endpoint terms are uniformly negligible. The trace-class covariance gives a Radon Gaussian law on the separable Hilbert space, completing tightness and identification. $\square$

Theorem 1.8 (Round-twenty-one A1 closure). *The benchmark admits a proper exact autonomous Hamiltonian suspension, an incidence-completed labelled current calculus with a fixed-fibre closed material derivative, a convergent all-order response formula, and a trace-class current-valued functional CLT. Each construction is made in the active manuscript and none uses restriction of an arbitrary negative Sobolev distribution.*

Proof. Combine Theorems 1.2, 1.3, 1.5, and 1.7 with Proposition 1.4. $\square$

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Twenty-One dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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